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Do suppliers value customer firms' digital transformation? Evidence from trade credit provision

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ABSTRACT

This study explores suppliers' attitudes towards customer firms' digital transformation and its consequences from the perspective of trade credit. It finds that Chinese public firms with a higher level of digital transformation receive more trade credit from suppliers. Mechanism tests show that digital transformation improves firms' information environment and technical efficiency while reducing financial performance volatility. This, in turn, decreases the information asymmetry and default risks suppliers face, enhancing firms' access to trade credit from suppliers. Further analysis reveals that this effect is stronger in firms with larger sizes, higher market shares, stronger profitability, and lower bankruptcy risks and is more pronounced for firms headquartered in provinces with lower levels of social trust or weaker law enforcement. Finally, we highlight the economic outcomes of this effect by showing that receiving more trade credit helps firms alleviate financing constraints, reduce debt costs, and improve investment efficiency.

1. Introduction

The digital economy is acting as a key force in reorganizing global resource elements, reshaping the global economic structure, and altering the global competitive landscape (Verhoef et al., 2021). Consequently, firms globally are progressively engaging in the application of digital technologies to revamp their production, operational, and management, i.e., digital transformation, which recently attracted notable academic attention (Broekhuizen, Broekhuis, et al., 2021; Nambisan et al., 2019; Vial, 2019). Extant studies focus on exploring factors driving firms' digital transformation (AlNuaimi et al., 2022; Du et al., 2023; Nambisan et al., 2017) and its impact on firms' stock market or financial performance (Chen & Srinivasan, 2024; Liu, Zhao, & Kong, 2023; Matarazzo et al., 2021). In contrast, the question of how external stakeholders respond to firms' digital transformation, the answer to which offers meaningful insights into the understanding of digital transformation's outcomes and their driving mechanisms, remains underexplored.

This study seeks to address this gap by empirically studying the impact of firms' digital transformation on a specific group of stakeholders, suppliers, from the perspective of their trade credit provision. Trade credit provided by suppliers is globally an important informal financing tool, especially in countries with underdeveloped formal financing markets such as banks and stock markets (Levine et al., 2018; Li et al., 2021). On the one hand, suppliers face significant risks in providing trade credit due to the absence of collateral or third-party guarantees behind the transaction (Wu et al., 2014). Accordingly, suppliers' willingness to provide trade credit reflects their attitudes to customer firms' digital transformation. On the other hand, suppliers possess informational advantages in assessing the prospects of their customers via routine transactions over time (Petersen & Rajan, 1997). Thus, the relationship between firms' digital transformation and their access to trade credit from suppliers provides valuable implications for the causes and potential outcomes of such transformations.

Like banks and other capital providers, suppliers primarily assess customer firms' credibility based on their information asymmetry risk and financial solvency when providing trade credit (Fabbri & Menichini, 2010; Gryglewicz, 2011). Naturally, suppliers are less willing to provide trade credit to firms with higher information asymmetry risk or default risk. Prior studies prove that digital transformation considerably relates to firms' information environment and financial performance (Chen et al., 2022; Liu, Wu, et al., 2023; Matarazzo et al., 2021). In other

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words, the firm's degree of digital transformation can be a crucial determinant of its information asymmetry risk and financial solvency, thereby determining its access to trade credit from suppliers.

We construct a sample of Chinese public firms from 2008 to 2022 to test the above prediction empirically. The Chinese setting is suitable for two reasons. First, China has explicitly proposed digital transformation as its national strategy in recent years. As of 2022, China's digital economy reached a scale of 50.2 trillion yuan, accounting for 41.5 % of its GDP.[2] Thus, digital transformation has emerged as a common strategy among Chinese firms, offering a unique setting for exploring its outcomes. Second, the formal financing markets in China, such as banks and stock markets, are still developing. Compared to developed countries, firms in China rely notably more on trade credit as an informal financing tool (Kong et al., 2020; Ouyang et al., 2024; Wu et al., 2014). Therefore, the expected influence of digital transformation on firms' capacity to secure trade credit should be more distinctly observable in China.

Following most prior studies on firm-level digital transformation (e.g., Chen and Srinivasan (2024) and Liu, Zhao, and Kong (2023)), we measure firms' digital transformation based on the number of digital transformation-related words mentioned in their annual reports (denoted as $DigTran_{i,t}$). Our empirical results show that firms with higher degrees of digital transformation receive significantly more trade credit from suppliers. All else equal, a one-standard-deviation increase in $DigTran_{i,t}$ leads to a 2.19 % increase in the firm's received trade credit (measured as its accounts payable to total liabilities), which equals a 9.16 % increase relative to the average trade credit of our sample firms. The positive impact of digital transformation on firms' access to trade credit is robust to alternative trade credit proxies and digital transformation proxies and still holds when firm-level fixed effects are included to address the concern about omitted variables.

We establish the causal direction of this impact by instrumenting $DigTran_{i,t}$ with the interaction term between the number of post offices per million people in 1984 in the city where firm i is headquartered ($Post_{i,1984}$) and the number of internet users per million households in that city for year t ($Internet_{i,t}$).[3] Better internet infrastructure facilitates local firms' digital transformation. Meanwhile, post offices predominantly installed landline telephones offering internet access via dial-up connections in early China. Thus, the density of post offices in a Chinese city's early years causes exogenous variation in its subsequent internet infrastructure. In contrast, the firm-level digital transformation cannot affect the city's postal office density in 1984 or the current city-level internet infrastructure, making $Post_{i,1984} \times Internet_{i,t}$ a valid instrumental variable. The two-stage least square (2SLS) regressions show that the instrumented $DigTran_{i,t}$ still significantly positively impacts firms' access to trade credit, suggesting that the impact is causal.

Next, we verify whether digital transformation enhances the firm's access to trade credit by reducing its information asymmetry risk and improving its financial solvency. As evidence, we find that digital transformation enhances firms' information environment, as indicated by reduced analyst earnings forecast errors and dispersion. It also helps improve financial solvency by increasing firms' total factor productivity and reducing their ROA volatility. Consistent with those mechanisms, our cross-sectional analyses show that the impact of digital transformation on trade credit is larger in firms with larger sizes, higher market shares, stronger profitability, and lower bankruptcy risks and is also greater for firms located in provinces with lower levels of social trust or weaker law enforcement. Finally, we highlight the economic consequence of the impact by showing that more available trade credit from suppliers helps Chinese firms alleviate financing constraints, reduce debt costs, and improve investment efficiency.

Our study is closely related to Liu and Wang (2023), Zhou and Li (2023), Zhang et al. (2023), and Cheng (2024), which also explore the impact of corporate digital transformation on trade credit financing in China. However, our study notably differs from these works regarding research questions, empirical design, findings, and theoretical basis. While Liu and Wang (2023) and Cheng (2024) examine the impact of digital transformation on firms' policies for providing trade credit to clients, we focus on how such transformation affects their access to trade credit from suppliers. Although Zhou and Li (2023) and Zhang et al. (2023) address the same research question as ours, they reach opposite conclusions, highlighting the ongoing debate about the relationship between corporate digital transformation and trade credit access. Our findings, which align with Zhang et al. (2023) but contradict Zhou and Li (2023), provide new empirical evidence and insights into this debate. Additionally, our exploration of the mechanisms and economic consequences of this relationship is unique compared to Zhang et al. (2023). While Zhang et al. (2023) rely on buyer market theory to explain the positive impact of digital transformation on trade credit access, our study adopts a creditor risk aversion perspective, arguing that digital transformation enhances firms' access to trade credit by reducing information asymmetry and default risks suppliers face. Thus, besides complementing Zhang et al. (2023), our study makes unique contributions to studies on corporate digital transformation and trade credit financing.

More specifically, we may contribute to the literature in the following ways. First, we shed light on the perspectives of suppliers, an important group of firm's external stakeholders, regarding firms' digital transformation by demonstrating its impact on their willingness to provide trade credit. While prior research has primarily focused on the effects of digital transformation on stock market performance (Jiang et al., 2022; Liu, Zhao, & Kong, 2023) or on operational and financial outcomes (Chen & Hao, 2022; Chen & Srinivasan, 2024), the role of external stakeholders has received less attention. In contrast, our findings show that suppliers react positively to firms' digital transformation efforts by extending more trade credit to firms with higher levels of digitalization. This enriches the existing literature by providing new insights into the role of stakeholders in the outcomes of digital transformation.

Second, we contribute to the understanding of the factors influencing firms' access to trade credit, a topic that has garnered significant academic interest given the importance of trade credit as a critical informal financing source, especially in developing countries with less developed formal financial markets (Levine et al., 2018). Existing research identifies several determinants of trade credit provision by suppliers, such as product characteristics (Giannetti et al., 2011), financial reporting quality (Li et al., 2021), financial health (Abdulla et al., 2020), national culture (El Ghoul & Zheng, 2016), and monetary policies (Adelino et al., 2022). We extend this body of work by highlighting that digitalization, a strategic transformation adopted globally in recent years, significantly influences firms' ability to secure trade credit. Our findings offer valuable implications for firms' financing strategies under digital transformation.

Finally, we add to the research on the factors that determine the outcomes of corporate digital transformation and the channels through which these outcomes are realized. Previous studies suggest that digital transformation can enhance productivity (Wang, 2023), boost innovation efficiency (Nambisan et al., 2017), and ultimately increase firm value (Berg et al., 2020). Our research reveals that more available trade credit due to digital transformation helps firms mitigate financing constraints, lower debt costs, and improve investment efficiency. These results indicate that suppliers' attitudes play a crucial role in the success of firms' digital transformation, with trade credit provision serving as a key channel for this effect. Thus, our findings provide fresh insights into how digital transformation influences operational performance and

[2] See http://www.caict.ac.cn/kxyj/qwfb/bps/202304/P020230427572038320317.pdf for the China Digital Economy Development White Paper (2023).

[3] As discussed in detail in Section 5.3.3, the interaction term is a Bartik instrument variable based on the methodology of Goldsmith-Pinkham et al. (2020).

offer practical implications for firms navigating their digital transformation process.

We organize the remainder of this study as follows: Section 2 reviews prior studies on digital transformation and trade credit. In Section 3, we propose testable hypotheses on the impact of digital transformation on trade credit. Section 4 introduces our sample, data, and baseline regression model, and Section 5 reports the baseline regression results and robustness tests. Section 6 conducts further analyses to verify how digital transformation impacts trade credit. Moreover, it discusses the cross-sectional variation in the impact and its economic outcomes. Finally, Section 7 concludes.

2. Literature review

2.1. Digital transformation

The swift evolution of digital technologies has significantly elevated the degree of digitization in both economic and social realms, thereby fundamentally altering economic landscapes and business paradigms (Verhoef et al., 2021). In response to the advent of the digital economy era, firms globally are increasingly undertaking digital transformation, that is, transforming production, operational, and management models through digital technologies and upgrading these models with digital solutions. Such digital transformation has attracted significant academic attention in the recent decade (Broekhuizen, Broekhuis, et al., 2021; Nambisan et al., 2019).

Existing research primarily investigates the factors determining firms' digital transformation and the outcomes of such transformation. In terms of factors driving transformation, extant studies broadly discuss factors related to macro-level social and economic conditions and micro-level firm characteristics. More specifically, a better business environment (Luo et al., 2023), more severe air pollution (Lu et al., 2024), or stronger intellectual property protection (Wen & Deng, 2023) motivate firms' digital transformation. In contrast, the expansion of local government debt inhibits such transformation by intensifying corporate financing constraints and distorting corporate investment preferences (Yang et al., 2023). For firm characteristics, Chen and Srinivasan (2024) show that American firms with larger sizes, higher return volatility, or more selling, general, and administrative expenses engage in more digital activities. Moreover, Chinese firms are associated with higher degrees of digital transformation if they have less concentrated customers (Li, Jiang, & Ma, 2023), stronger non-family shareholder governance (Du et al., 2023), or if their customers' levels of digital transformation are higher (Guo et al., 2023).

Most prior studies affirm the positive outcomes of digital transformation. On the one hand, digital transformation can enhance a firm's operational efficiency and productivity by innovating business models, creating new distribution channels or ways to create and deliver value to customers (Levinsohn & Petrin, 2003; Matarazzo et al., 2021; Wang, 2023). On the other hand, the connective, shared, and open nature of digital transformation helps reduce information asymmetry (Chen et al., 2022), thereby exerting a governance effect on managers (Huang et al., 2023). The improvements in operational efficiency and governance increase firms' effectiveness of internal controls (Zhao et al., 2023), optimize managers' risk-taking levels (Kong & Liu, 2023), and, thus, enhance firms' stock and financial performance (Jiang et al., 2022; Wu et al., 2022). Also, digital transformation improves firms' non-financial performance, such as enhancing environmental sustainability (Chen & Hao, 2022), better fulfilling corporate social responsibilities (Kong & Liu, 2023), improving employee rights (Li, Liu, et al., 2023), and motivating innovation (Yu & Shao, 2024).

Nevertheless, several studies suggest that digital transformation does not necessarily lead to positive outcomes and may even result in adverse consequences. The cost of utilizing digital technology to replace and upgrade old equipment and reshape business processes and operational models is substantial. Meanwhile, the benefits of digital transformation may take considerable time to materialize (Brynjolfsson et al., 2019). Moreover, digital transformation requires developing complementary organizational capabilities. The potential benefits of digital transformation may be offset by the lack of such capabilities (Chen & Srinivasan, 2024).

As evidence supporting these arguments, Liu, Zhao, and Kong (2023) and Huang et al. (2023) show that digital transformation may amplify instead of mitigate stock price volatility under certain conditions, while Tian et al. (2022) suggest that digital transformation amplifies corporate risk-taking for Chinese firms. Also, while digital finance may increase financial inclusion and upgrade consumption (Zhou et al., 2023), it may also heighten the risk of households falling into a debt trap (Yu & Shao, 2024). Finally, Hao et al. (2023) demonstrate that the advancement of digital finance adversely affects the ability of banks to create liquidity.

2.2. Trade credit

Trade credit refers to the practice where suppliers allow customers to delay payment for goods. Trade credit occupies a significant position in global commercial activities, particularly in countries where formal financial markets are underdeveloped (Li et al., 2021). In the 34 countries surveyed by Levine et al. (2018), the average ratio of trade credit to firms' total liabilities was 25 %. The use of trade credit can alleviate a firm's financing constraints, thereby assisting firms in navigating financial crises, responding to monetary policy tightening (Biais & Gollier, 1997; McGuinness et al., 2018), reducing stock price crash risks (Cao et al., 2018), or enhancing technical efficiency (Agostino & Trivieri, 2019). Compared to developed countries, China's formal financial markets, such as banks, are less developed. Thus, Chinese firms, especially private firms, rely more heavily on trade credit as an informal financing tool (Allen et al., 2005; Kong et al., 2020).

Given the importance of trade credit, extensive studies have explored the determinants of a firm's ability to obtain trade credit. These studies mainly revolve around two categories of theories: the substitutionary financing theory (i.e., the financing motivation) and the buyer market theory (i.e., the transaction motivation). The substitutionary financing theory views trade credit as an alternative financing tool when formal financing channels like bank credit are constrained. Hence, a firm's use of trade credit depends on its financing needs and the availability of formal financing channels. For instance, Petersen and Rajan (1997) note that smaller firms with lower credit quality, having limited access to capital markets or bank credit, tend to use trade credit more. Levine et al. (2018) find that trade credit is more utilized in countries with underdeveloped financial markets, thus relying more on its substitutionary financing role. Using the liberalization of the upper and lower limits of bank interest rates in China as an external shock to bank credit availability, Chen, Ma, and Wu (2019) find a significant negative relationship between the amount of trade credit used by firms and their availability of bank credit.

On the other hand, the buyer market theory posits that suppliers offer trade credit to customers out of transaction motives, using it as a means to enhance product competitiveness, establish long-term relationships with customers, and expand sales. First, suppliers agreeing to delayed payment after the delivery of goods demonstrate their confidence in the quality of the goods, which can alleviate customers' concerns about inferior quality, thereby achieving expanded sales (Lehar et al., 2020). Secondly, trade credit often includes terms for discounts on early payments, and customers choosing to pay upon maturity pay higher prices for the same goods. Thus, suppliers use trade credit to achieve price discrimination among customers, capturing more surplus value (El Ghoul & Zheng, 2016). Finally, the growth of a customer leads to increased demand for intermediate goods, from which the supplier benefits. This relationship implies that the supplier implicitly holds equity interests in their customer firms, hence motivated to provide trade credit to ensure their customers' liquidity (Cunat, 2007; Dass et al., 2015).

3. Research hypotheses

As shown in Section 2, despite the abundance of research examining the determinants and potential consequences of corporate digital transformation, studies from the perspective of suppliers remain scarce. To reveal suppliers' attitudes towards their customer firms' digital transformation, we study the impact of digital transformation on firms' access to trade credit.

Similar to banks' assessment of credit risk, the willingness of suppliers to provide trade credit depends on the credibility of their customer firms. Information asymmetry and financial solvency (i.e., debt fulfillment capacity) are two primary determinants of a firm's credibility (Fabbri & Menichini, 2010). In an environment of imperfect information, suppliers might misjudge a customer firm's credibility due to information asymmetry. Moreover, the firm might fail to fulfill credit repayment obligations due to poor financial performance, leading to default risk.

Existing studies have shown that digital transformation significantly affects both firms' information environment and financial performance. In terms of the information environment, digital technologies are characterized by connectivity, sharing, and openness (Broekhuizen, Emrich, et al., 2021), which help firms better process and disseminate information, mitigating internal and external information asymmetries. Empirical research has confirmed the role of digital transformation in alleviating information asymmetry, evidenced by increased analyst coverage, enhanced public information accuracy, and reduced stock price crash risk (Chen et al., 2022; Wu et al., 2022).

Additionally, most research suggests that digital transformation contributes to improved operational efficiency and market value, thereby enhancing a firm's financial solvency. For instance, the case studies of Matarazzo et al. (2021) show that digital transformation helps small and medium-sized Italian firms create new distribution channels and ways to create and deliver value to customer segments. Liu, Wu, et al. (2023) reveal that digital transformation considerably fine-tunes resource distribution and enhances labor investment efficiency for Chinese firms, while Wang (2023) suggests that digital transformation increases the total factor productivity for Chinese firms by enhancing corporate innovation.

In summary, we expect that digital transformation reduces firms' information asymmetry and improves their operational performance, thereby decreasing the information asymmetry and default risks faced by suppliers when providing trade credit. They then may provide more trade credit to firms with higher degrees of digital transformation. Based on these assertations, we propose the following hypothesis:

Hypothesis H1. Firms with a higher degree of digital transformation receive more trade credit from suppliers.

The impact of digital transformation on trade credit may vary with firm size and market share. On the one hand, as a strategic transformation, digitalization has a long-term and uncertain nature (Brynjolfsson et al., 2019). To materialize the expected long-term benefits of digitalization, firms must possess adequate financial resources to afford the high initial costs of digital transformation and the capacity to bear the risks in the throes of transition (Chen & Srinivasan, 2024). On the other hand, due to the low marginal cost and significant scale effects of digital technologies, the benefits of digital transformation in reducing transaction costs, enhancing resource utilization efficiency, and promoting sales expansion become evident only after surpassing a certain threshold (Hajli et al., 2015).

Compared to small and medium-sized enterprises, larger firms with more financial resources and stronger risk resistance capabilities can ensure sustained and substantial investment for digital transformation and benefit more from scale effects. Similarly, firms with larger market shares are usually industry leaders enjoying higher market positions, thereby having greater control over industry resources and bargaining power with upstream and downstream enterprises. They are better equipped to withstand the transformation risk of digitalization and more likely to meet the production and sales thresholds for the scale effects of digital transformation. Therefore, the positive impact of digital transformation on operational performance should be stronger in larger firms or firms with higher market shares. Consequently, suppliers' responses to digital transformation in such firms might be more favorable. That is, the impact of digital transformation on firms' access to trade credit from suppliers should be greater for such firms. Based on these arguments, we propose the following hypothesis:

Hypothesis H2. The positive impact of digital transformation on trade credit access is greater in firms with larger sizes or higher market shares.

Prior studies suggest that various frictions may occur when applying new technologies. Examples of such frictions include the requirement of developing complementary organizational capabilities and the high fixed costs of implementation and creating new markets (Dranove et al., 2014). If a firm already faces high operational and financial risks, the frictions of digital transformation might exacerbate the fluctuation of the firm's operational and financial performance, leading to high default risks for suppliers. Given this, the higher a firm's existing operational and financial risks, the more cautiously suppliers might treat its digital transformation, suggesting a weaker effect of digital transformation in enhancing the access to trade credit for such firms.

High profitability indicates sound financial status and relatively low default risk, while high financial risk increases the likelihood of bankruptcy, making bankruptcy risk an important proxy for financial risk. Hence, we anticipate that a firm's profitability and bankruptcy risk affect the enhancement effect of digital transformation on its access to trade credit. We then propose the following hypothesis:

Hypothesis H3. The positive impact of digital transformation on trade credit access is larger in firms with higher profitability and lower bankruptcy risk.

Finally, the impact of digital transformation on trade credit may vary in regions. We argue that digital transformation enhances the perceived trustworthiness of suppliers on customer firms, thereby increasing firms' access to trade credit from suppliers. Prior studies show that local social trust levels matter to the degree of trust between suppliers and customer firms and, therefore, affect suppliers' provision of trade credit (Kong et al., 2020; Wu et al., 2014). Based on these studies, we expect a lower marginal enhancement effect of digital transformation on firms' trustworthiness for firms located in regions with higher local social trust levels. If so, the impact of digital transformation on trade credit should be weaker for such firms.

Moreover, trade credit is essentially a debt contract between suppliers and customers, and the legal environment determines the quality of contract fulfillment (Foroughi et al., 2022). In regions with effectively enforced legal systems, customers face higher legal costs defaulting on trade credit payments, and the interests of suppliers as creditors are better legally protected, reducing their risk in providing trade credit. Thus, a good legal environment and strong law enforcement can partially substitute for the increase in the firm's trustworthiness resulting from its digital transformation. Consequently, we expect that the positive impact of digital transformation on trade credit access should be relatively weaker in regions with stronger law enforcement.

Based on the above discussions, we propose the following hypothesis:

Hypothesis H4. The positive impact of digital transformation on trade credit access is weaker in regions with higher social trust levels and stronger law enforcement.

4. Research design

4.1. Baseline regression model

We construct the following model to empirically test the impact of firms' digital transformation on their access to trade credit:

$$TCredit_{i,t+1} = \alpha + \beta_1 DigTran_{i,t} + \sum Controls + Industry\ FE + Year\ FE + \varepsilon_{i,t} \quad (1)$$

where $TCredit_{i,t+1}$ is the dependent variable proxying for firm i's access to trade credit in year $t+1$. Following prior studies on the Chinese A-share market (e.g., Wu et al. (2014), Chen, Ma, and Wu (2019), and Kong et al. (2020)), we define $TCredit_{i,t+1}$ as firm i's ratio of accounts payable to its total liabilities. The explanatory variable of most interest in Eq. (1) is $DigTran_{i,t}$, which proxies for firm i's degree of digital transformation in year t. We use the one-year leading trade credit proxy in Eq. (1) to mitigate the concern about the bidirectional or reverse causality between the firm's digital transformation and its access to trade credit.

Existing studies commonly use the number of digital transformation-related words mentioned in the firm's annual report to capture its degree of digital transformation (Chen & Srinivasan, 2024; Guo et al., 2023; Kong & Liu, 2023). The idea is that digitalization is considered a crucial transformation strategy, and firms typically summarize their strategies in annual reports. In line with prior studies, we define $DigTran_{i,t}$ as the logarithm of one plus the number of digital transformation-related words mentioned in firm i's annual report of year t. Specifically, these words encompass terms associated with prevalent digital technologies such as "artificial intelligence," "blockchain," "cloud computing," and "big data technologies," as well as those associated with the application of these technologies. A comprehensive list of those words is provided in the Online Appendix A. The China Stock Market & Accounting Research (CSMAR) Database tracks and records the number of such words in each Chinese public firm's annual reports. We obtain these records to construct $DigTran_{i,t}$.

Following extant studies, we add a set of known determinants of trade credit as control variables in Eq. (1), including firm i's size ($Size_{i,t}$), book-to-market ratio ($BM_{i,t}$), bank loans ($BCredit_{i,t}$), short-term debt ($SDebt_{i,t}$), total liabilities ($Lev_{i,t}$), cash holdings ($Cash_{i,t}$), the proportion of tangible assets ($Tan_{i,t}$), return on assets ($ROA_{i,t}$), sales growth rate ($SaleGr_{i,t}$), and the indicator of state-owned-enterprise ($SOE_{i,t}$). Detailed definitions for each variable can be found in the Appendix of this study.

We control for those variables for the following reasons: First, smaller firms face greater difficulties in formal financial markets, making them more reliant on bridging funding gaps through trade credit (Kong et al., 2020). Secondly, trade credit serves as an important supplement to formal credit tools such as bank loans. Therefore, the accessibility of bank loans is a crucial determinant of a firm's trade credit demand (Chen, Harford, & Kamara, 2019). Additionally, financial metrics such as short-term debt, cash holdings, total debt, and tangible asset ratios relate to a firm's ability to meet debt obligations and, thus, determine its access to trade credit. Furthermore, return on assets, sales growth, and book-to-market ratios reflect the firm's operational and investment prospects, influencing suppliers' willingness to grant trade credit (Quan et al., 2024). Finally, compared to state-owned enterprises (SOEs), private firms in China rely more on trade credit as an informal financing tool since the Chinese banking sector is predominantly state-controlled and shows a bias towards SOEs when granting loans (Cong et al., 2019).

4.2. Data and sample

In 2007, China undertook major reforms to its accounting standards to align them with the International Financial Reporting Standards (IFRS). Considering this, we start our sample from 2008 and use Chinese public firms spanning 2008–2022 as our initial sample. We then drop observations from firms in the financial industry and those specially treated (ST firms) due to their financial distress or delisting risk. We also refer to the 2012 industry classification standards issued by the China Securities Regulatory Commission (CSRC) and exclude firms in industries such as 'Computer, Communications, and Other Electronic Equipment Manufacturing,' 'Software and Information Technology Services,' 'Internet and Related Services,' and 'Telecommunications, Broadcasting, Television, and Satellite Transmission Services.' For firms in these industries, digital technologies may serve as necessary production means rather than a tool for strategic transformation. Hence, it is necessary to exclude those industries from the sample.[4] Finally, we drop observations with missing control variables. Our final sample consists of a maximum of 25,498 observations obtained from 3268 unique Chinese public firms. Unless otherwise stated, all other data is sourced from CSMAR.

Panel A of Table 1 reports the descriptive statistics for the main variables. To mitigate the impact of extreme outliers, we winsorize all

Table 1
Descriptive Statistics and Univariate Sorting Analysis.

Panel A Descriptive Statistics for the Full Sample

Variables	# of Obs	Mean	SD	P25	Median	P75
$TCredit_{i,t+1}$	25,498	0.239	0.165	0.111	0.202	0.334
$DigTran_{i,t}$	25,498	1.034	1.290	0.000	0.693	1.792
$Size_{i,t}$	25,498	1.392	1.289	0.459	1.218	2.134
$BM_{i,t}$	25,498	0.622	0.244	0.434	0.623	0.811
$SOE_{i,t}$	25,498	0.429	0.495	0.000	0.000	1.000
$BCredit_{i,t}$	25,498	0.295	0.230	0.072	0.289	0.470
$SDebt_{i,t}$	25,498	0.582	0.187	0.452	0.591	0.721
$Lev_{i,t}$	25,498	0.442	0.211	0.273	0.436	0.600
$Cash_{i,t}$	25,498	0.179	0.127	0.089	0.144	0.231
$Tan_{i,t}$	25,498	0.927	0.089	0.915	0.956	0.979
$ROA_{i,t}$	25,498	0.037	0.061	0.014	0.036	0.066
$SaleGr_{i,t}$	25,498	0.191	0.464	−0.016	0.112	0.276

Panel B Univariate Sorting Analysis

	Low $DigTran_{i,t}$		*High* $DigTran_{i,t}$		*Low* minus *High*	
	# of Obs	Mean	# of Obs	Mean	Mean differences	T-statistics
$TCredit_{i,t+1}$	12,307	0.216	13,191	0.253	−0.037	−18.43***
$Size_{i,t}$	12,307	1.260	13,191	1.516	−0.256	−15.92***
$BM_{i,t}$	12,307	0.635	13,191	0.609	0.026	8.42***
$SOE_{i,t}$	12,307	0.501	13,191	0.363	0.138	22.51***
$BCredit_{i,t}$	12,307	0.336	13,191	0.257	0.079	27.82***
$SDebt_{i,t}$	12,307	0.588	13,191	0.576	0.012	5.27***
$Lev_{i,t}$	12,307	0.463	13,191	0.422	0.041	15.49***
$Cash_{i,t}$	12,307	0.168	13,191	0.189	−0.021	−13.10***
$Tan_{i,t}$	12,307	0.937	13,191	0.917	0.020	18.30***
$ROA_{i,t}$	12,307	0.034	13,191	0.040	−0.005	−6.69***
$SaleGr_{i,t}$	12,307	0.185	13,191	0.197	−0.012	−2.03**

Panel A of this table reports the number of observations, mean, standard deviation, 25th percentile, median, and 75th percentile for each variable. Panel B divides the sample into two groups based on whether $DigTran_{i,t}$ is above or below the sample median. It then compares the mean differences in $TCredit_{i,t+1}$ and other variables between the two groups. $DigTran_{i,t}$ is the logarithm of one plus the number of digital transformation-related feature words mentioned in firm i's annual report of year t. $TCredit_{i,t+1}$ is the ratio of firm i' accounts payable to its total liabilities at the end of year $t+1$. Detailed definitions for each variable are provided in the Appendix. *, **, and *** denote significance at the 10 %, 5 %, and 1 % levels, respectively.

[4] Our main findings do not qualitatively change when we do not exclude those industries from the sample.

continuous variables at the 1 % and 99 % levels. The mean of $TCredit_{i,t+1}$ indicates that accounts payable constitute, on average, 23.9 % of total liabilities for our sample firms, highlighting the significant role of trade credit for Chinese firms. The mean of $DigTran_{i,t}$ is 1.034 and is comparable to other studies on digital transformation-related topics in China (e.g., Guo et al. (2023) and Kong and Liu (2023)). Moreover, the correlation matrix provided in the Online Appendix B shows that $DigTran_{i,t}$ is significantly and positively correlated with $TCredit_{i,t+1}$, providing preliminary evidence supporting Hypothesis H1 that a firm's digital transformation enhances its access to trade credit from suppliers.

5. Empirical results

5.1. Univariate analysis

First, we explore the relationship between firms' digital transformation and their access to trade credit through univariate analysis. In Panel B of Table 1, we divide observations into two groups based on whether a firm's degree of digital transformation ($DigTran_{i,t}$) is above the sample median. This division allows for a comparison of trade credit amounts and firm characteristics between the two firm groups. In the group with a lower $DigTran_{i,t}$, the average proportion of accounts payable to total liabilities is 21.6 %, whereas this ratio is 25.3 % in the group with a higher $DigTran_{i,t}$. The difference of 3.7 % is statistically significant (t-statistic = −18.43), consistent with the prediction of Hypothesis H1 that firms with higher degrees of digital transformation receive more trade credit. However, Panel B also reports significant differences in firm characteristics between the two firm groups. Therefore, we employ multivariate regressions in the following Section to formally verify the impact of digital transformation on trade credit while controlling for the firm-level characteristics.

5.2. Baseline regression results

Table 2 presents the baseline regression results from estimating Eq. (1). In all regressions, we control for industry and year fixed effects and use robust standard errors to infer the statistical significance of coefficients. In Column (1), we regress the firm's trade credit proxy ($TCredit_{i,t+1}$) on its digital transformation proxy ($DigTran_{i,t}$) and the industry and year fixed effects. The coefficient of $DigTran_{i,t}$ is positive (0.017) and highly significant (t-value =17.70). We then gradually include control variables in Columns (2) to (4), where the coefficients of $DigTran_{i,t}$ remain positive and highly significant. For instance, after incorporating all control variables in Column (4), the coefficient of $DigTran_{i,t}$ is 0.014 (t-value =18.64). The coefficient implies that for a one-standard-deviation increase (1.290) in a firm's digital transformation, its trade credit obtained from suppliers increases by (1.290 × 0.017) = 2.19 %. This impact is also economically significant since the 2.19 % increase equals a 0.0219/0.239 = 9.16 % increase relative to the sample mean (0.239) of $TCredit_{i,t+1}$. Overall, the results in Table 2 support Hypothesis H1 and suggest that firms with higher levels of digital transformation secure significantly more trade credit from suppliers, indicating that suppliers hold a positive attitude towards their customer firms' digital transformation.[5]

The results of the control variables are largely consistent with existing research. The coefficients for $Size_{i,t}$ and $SOE_{i,t}$ are significantly negative, consistent with the notion that smaller and non-SOEs in China are more dependent on trade credit to fill funding gaps since they are more likely to face discrimination in the formal financial system, such as banks (Allen et al., 2005; Cong et al., 2019). $BCredit_{i,t}$ is significantly negatively correlated with $TCredit_{i,t+1}$, aligning with existing research that finds a substitution relationship between trade credit and bank loans (Chen, Harford, & Kamara, 2019; Norden et al., 2020). Both $SDebt_{i,t}$ and $Lev_{i,t}$ have significantly negative coefficients. On the one hand, there is a substitution relationship between trade credit and other debt financing tools. On the other hand, higher current or total liabilities imply a greater financial risk, reducing suppliers' willingness to offer trade credit. Conversely, the coefficients of the book-to-market ratio ($BM_{i,t}$) and the proportion of tangible assets ($Tan_{i,t}$) are significantly positive. Firms with higher book values and a greater proportion of tangible assets have more assets available for collateral. Thus, suppliers are more willing to grant those firms trade credit because of their stronger debt-payment capacity (Kong et al., 2020; Ouyang et al., 2024). Finally, the firm's cash holdings ($Cash_{i,t}$) and return on assets ($ROA_{i,t}$) show no significant statistical correlation with $TCredit_{i,t+1}$.

Table 2
Baseline regression results.

	(1)	(2)	(3)	(4)
	$TCredit_{i,t+1}$	$TCredit_{i,t+1}$	$TCredit_{i,t+1}$	$TCredit_{i,t+1}$
$DigTran_{i,t}$	0.017***	0.019***	0.013***	0.014***
	(17.70)	(20.04)	(16.78)	(18.64)
$Size_{i,t}$		−0.025***	−0.022***	−0.022***
		(−26.12)	(−26.12)	(−25.05)
$BM_{i,t}$		0.015***	0.043***	0.044***
		(2.76)	(9.34)	(9.39)
$SOE_{i,t}$			0.006***	0.003*
			(3.39)	(1.86)
$BCredit_{i,t}$			−0.230***	−0.225***
			(−58.89)	(−58.09)
$SDebt_{i,t}$			−0.308***	−0.306***
			(−59.43)	(−59.51)
$Lev_{i,t}$			−0.018***	−0.025***
			(−3.57)	(−4.47)
$Cash_{i,t}$			0.072***	0.045***
			(9.13)	(5.60)
$Tan_{i,t}$				0.183***
				(19.73)
$ROA_{i,t}$				0.035**
				(2.13)
$SaleGr_{i,t}$				−0.000
				(−0.19)
Constant	0.218***	0.241***	0.465***	0.299***
	(162.89)	(73.75)	(93.68)	(31.56)
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
F-statistics	313.313	432.867	1387.450	1047.232
Adjusted R^2	0.114	0.145	0.431	0.440
# of Obs	25,498	25,498	25,498	25,498

This table reports the baseline regression results obtained from estimating Eq. (1) in the text. The dependent variable, $TCredit_{i,t+1}$, is the ratio of firm i' accounts payable to its total liabilities at the end of year $t+1$. The explanatory variable of interest is $DigTran_{i,t}$, which is the logarithm of one plus the number of digital transformation-related feature words mentioned in firm i's annual report of year t. We gradually control for firm i's size ($Size_{i,t}$), book-to-market ratio ($BM_{i,t}$), the quantity of bank loans ($BCredit_{i,t}$), short-term debt ($SDebt_{i,t}$), total liabilities ($Lev_{i,t}$), cash holdings ($Cash_{i,t}$), proportion of tangible assets ($Tan_{i,t}$), return on assets ($ROA_{i,t}$), sales growth rate ($SaleGr_{i,t}$), and the indicator of state-owned-enterprise ($SOE_{i,t}$) in the regressions. Detailed definitions of each variable are provided in the Appendix. We include year fixed effects and industry fixed effects in all regressions. Parentheses contain t-statistics calculated using robust standard errors. *, **, and *** represent significance at the 10 %, 5 %, and 1 % levels, respectively.

[5] Our finding aligns with Zhang et al. (2023) but contradict Zhou and Li (2023). The different findings between this study and Zhou and Li (2023) may stem from differences in the sample, measurement of key variables, and regression model specifications. Moreover, although our findings align with those of Zhang et al. (2023), we notably differ from Zhang et al. (2023) in theoretical mechanisms and empirical design. Therefore, our study not only complements Zhang et al. (2023) but also makes unique contributions to studies on corporate digital transformation and trade credit financing.

5.3. Robustness tests

5.3.1. Alternative proxies for trade credit

The previous Section use the ratio of accounts payable to total liabilities as the proxy of trade credit. For robustness tests, this Section employs several alternative measures of trade credit following the idea of extant studies such as Cao et al. (2018), Kong et al. (2020), and Quan et al. (2024). Specifically, we use the ratio of accounts payable to total assets ($TCredit2_{i,t+1}$), the sum of accounts payable and notes payable as a proportion of total liabilities ($TCredit3_{i,t+1}$), or as a proportion of total assets ($TCredit4_{i,t+1}$) as alternative trade credit proxies. As shown in Table 3, regardless of whether control variables are included, the firm's digital transformation proxy ($DigTran_{i,t}$) is positively correlated with all three alternative trade credit proxies and remains significant at the 1 % level. These results suggest that the significant positive impact of digital transformation on firms' access to trade credit from suppliers is robust to different measures of trade credit.

5.3.2. Alternative proxies for digital transformation

In previous sections, we use the number of digital transformation-related words mentioned in the firm's annual report to proxy for its level of digital transformation. There are two concerns regarding the validity of this proxy. First, the number of digital transformation-related words mentioned in annual reports may depend on the total number of words used. In this way, $DigTran_{i,t}$ can be biased since larger firms may naturally use more words in annual reports to disclose their more complicated operations. To address this concern, we alternatively measure firms' digital transformation as the ratio of digital transformation-related words to the total number of words used in their annual reports and denote it as $PerWords_{i,t}$.

The other concern is that *DigTran* may only capture the firm's exposure to the digital economy or the extent of its voluntary disclosure regarding its digital transformation strategy rather than reflecting its degree of digital transformation. To address this concern, we alternatively measure the firm's degree of digital transformation as the ratio of its digitalization-related intangible assets to total assets. Chinese A-share public firms disclose the details of their intangible assets in the financial notes of their annual reports. We go through those notes to identify the digitalization-related intangible assets.[6] We then calculate their proportion of total assets to construct the aforementioned digital transformation proxy and denote it as $DigAsset_{i,t}$.[7]

Panel A (B) of Table 4 reports the results re-estimating Eq. (1) while using $PerWords_{i,t}$ ($DigAsset_{i,t}$) as the alternative proxy for digital transformation. In both Panels, Columns (1) to (4) use $TCredit_{i,t+1}$, $TCredit2_{i,t+1}$, $TCredit3_{i,t+1}$, and $TCredit4_{i,t+1}$ as dependent variables, respectively. The coefficients of $PerWords_{i,t}$ and $DigAsset_{i,t}$ are always positive with a 1 % significance level, confirming that the positive impact of digital transformation on trade credit does not suffer from potential measurement errors of digital transformation and, therefore, is robust.

5.3.3. Controlling for firm-fixed effects

Numerous studies have investigated the factors influencing firms' ability to obtain trade credit. Despite controlling for multiple firm characteristics in Eq. (1), there remains a possibility that the omission of important variables could affect the robustness of our results. To address this concern, we include firm fixed effects in the regression to deal with the potential issue of omitted variables. Table 5 presents the regression results with firm fixed effects. Columns (1) to (4) measure firms' trade credit with the ratio of accounts payable to total liabilities ($TCredit_{i,t+1}$), the ratio of accounts payable to total assets ($Tcredit2_{i,t+1}$), and the sum of accounts payable and notes payable, as the proportion of total liabilities or total assets ($Tcredit3_{i,t+1}$ and $Tcredit4_{i,t+1}$), respectively. Regardless of the trade credit proxy used, the coefficient of $DigTran_{i,t}$ is always positive at the 1 % significance level, indicating that the significant positive impact of digital transformation on trade credit is not affected by potentially omitted variables.

5.3.4. Establishing causality with the instrumental variable approach

So far, previous sections reveal a significant positive correlation between firms' digital transformation and the amount of their received trade credit. However, this correlation may be subject to endogeneity issues, such as bidirectional or reverse causality. For instance, firms face fewer financing constraints after they obtain more trade credit from suppliers. Those firms are then more capable of ensuring the capital required for digital transformation, thereby achieving a higher degree of digitalization. To address this endogeneity concern, we employ an instrumental variable approach and the two-stage least squares (2SLS) regressions to establish the causal direction between a firm's digital transformation and its access to trade credit. Specifically, we construct a Bartik instrument variable (IV) of $DigTran_{i,t}$ using the interaction term between $Post_{i,1984}$ and $Internet_{i,t}$.[8] $Post_{i,1984}$ is the number of post offices per million people in 1984 in the city where firm *i* is headquartered,[9] while $Internet_{i,t}$ is the number of internet users per million households in that city for year *t*.[10]

The Internet constitutes the cornerstone infrastructure of the digital economy. A larger number of internet users in a city indicates a more developed digital economic infrastructure conducive to facilitating local firms' digital transformation. Internet access historically relied on dial-up connections via landline telephones, and post offices predominantly installed these telephone lines in China. Consequently, the density of post offices in a Chinese city's early years leads to the exogenous variation in the subsequent intensity of its internet infrastructure. Meanwhile, a given firm's digital transformation can not influence the city's postal office density in 1984 or city-level internet infrastructure. Therefore, $Post_{i,1984} \times Internet_{i,t}$ satisfies the exogeneity requirement necessary for an instrumental variable.

Table 6 reports the 2SLS regression results using $Post_{i,1984} \times Internet_{i,t}$ as the instrumental variable. In Column (1), $Post_{i,1984} \times Internet_{i,t}$ is significantly positively correlated with $DigTran_{i,t}$, consistent with the expectation that a city's internet infrastructure facilitates local firms' digital transformation. The Cragg-Donald Wald F-statistic is 129.562, and the Kleibergen-Paap rk Wald F-statistic is 97.124. Both of them are much higher than the Stock-Yogo weak identification test critical value for a 10 % maximal IV size (16.38), rejecting the hypothesis that $Post_{i,1984} \times Internet_{i,t}$ is a weak instrumental variable. Additionally,

[6] Specifically, we define intangible assets related to "software," "networks," "client systems," "management systems," and "intelligent platforms," as disclosed in the firm's financial report notes, as digital technology-related intangible assets.

[7] To our knowledge, this proxy is not commonly used in existing relevant studies (e.g., Zhou and Li (2023), Zhang et al. (2023), Liu and Wang (2023), and Cheng (2024)). Accordingly, employing this proxy as an alternative indicator on firms' digital transformation may add to the novelty and uniqueness of our study.

[8] The Bartik instrumental variable, also known as the "shift-share" instrument, is a interaction term of the "shift" variable and the "share" variable. By providing a source of exogenous variation, the Bartik instrument can help in obtaining more unbiased and consistent estimates of causal effects. In our study, $Post_{i,1984}$ is the "share" variable and $Internet_{i,t}$ is the "shift variable". See Galindo-Martín et al. (2019) for details about the construction and advantages of the Bartik instrumental variable.

[9] We choose the year of 1984 due to the availability of the data on post offices in early China.

[10] Zhou and Li (2023) and Zhang et al. (2023) use the current internet infrastructure of the region where the firm is headquartered as an instrumental variable for its digital transformation. Considering the potential endogeneity between economic infrastructure and the external business environment of local firms, we argue that the Bartik instrumental variable we use is more exogenous than that employed by Zhou and Li (2023) and Zhang et al. (2023).

Table 3
Robustness Tests: Alternative trade credit proxies.

	(1)	(2)	(3)	(4)	(5)	(6)
	$TCredit2_{i,t+1}$	$TCredit2_{i,t+1}$	$TCredit3_{i,t+1}$	$TCredit3_{i,t+1}$	$TCredit4_{i,t+1}$	$TCredit4_{i,t+1}$
$DigTran_{i,t}$	0.006***	0.006***	0.017***	0.012***	0.006***	0.006***
	(15.41)	(19.17)	(17.43)	(15.34)	(12.89)	(14.98)
ΣControls	No	Yes	No	Yes	No	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
F-statistics	237.323	781.862	303.852	1607.175	166.243	1205.097
Adjusted R^2	0.132	0.421	0.089	0.475	0.143	0.526
# of Obs	25,498	25,498	25,498	25,498	25,498	25,498

This table reports the estimation results of Eq. (1) in the text while using alternative trade credit proxies as dependent variables. $DigTran_{i,t}$ is the logarithm of one plus the number of digital transformation-related words mentioned in firm *i*'s annual report of year *t*. $TCredit2_{i,t+1}$ is the ratio of firm *i*'s accounts payable to total assets at the end of year $t+1$. $TCredit3_{i,t+1}$ is the ratio of the sum of firm *i*'s accounts payable and notes payable to total liabilities, while $TCredit4_{i,t+1}$ is the ratio of the sum of firm *i*'s accounts payable and notes payable to total assets. ΣControls denotes the set of control variables used in Column (4) of Table 2. We include year fixed effects and industry fixed effects in all regressions. Parentheses contain t-statistics calculated using robust standard errors. *, **, and *** represent significance at the 10 %, 5 %, and 1 % levels, respectively.

Table 4
Robustness Tests: Alternative proxies for Digital Transformation.

Panel A Using $PerWords_{i,t}$ as the alternative proxy for digital transformation

	(1)	(2)	(3)	(4)
	$TCredit_{i,t+1}$	$TCredit2_{i,t+1}$	$TCredit3_{i,t+1}$	$TCredit4_{i,t+1}$
$PerWords_{i,t}$	0.818***	0.322***	0.742***	0.314***
	(9.21)	(7.24)	(9.35)	(8.22)
ΣControls	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
F-statistics	1002.067	731.184	1559.637	1165.262
Adjusted R^2	0.434	0.415	0.472	0.523
# of Obs	25,498	25,498	25,479	25,479

Panel B Using $DigAsset_{i,t}$ as the alternative proxy for digital transformation

	(1)	(2)	(3)	(4)
	$TCredit_{i,t+1}$	$TCredit2_{i,t+1}$	$TCredit3_{i,t+1}$	$TCredit4_{i,t+1}$
$DigAsset_{i,t}$	0.005***	0.003***	0.005***	0.003***
	(4.48)	(7.45)	(4.23)	(5.70)
ΣControls	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
F-statistics	801.249	561.318	1226.318	846.678
Adjusted R^2	0.441	0.418	0.477	0.518
# of Obs	19,305	19,305	19,305	19,305

This table reports the estimation results of Eq. (1) in the text while using alternative proxies for corporate digital transformation. $PerWords_{i,t}$ is the percentage of digital transformation-related words to the total number of words used in firm *i*'s annual report of year *t*. $DigAsset_{i,t}$ denotes the ratio of firm *i*'s digital technology-related intangible assets to its total assets. ΣControls denotes the set of control variables used in Column (4) of Table 2. We include year fixed effects and industry fixed effects in all regressions. Parentheses contain t-statistics calculated using robust standard errors. *, **, and *** represent significance at the 10 %, 5 %, and 1 % levels, respectively.

Column (2) shows that the digital transformation proxy instrumented by $Post_{i,1984} \times Internet_{i,t}$ is significantly positively correlated with the trade credit proxy ($TCredit_{i,t+1}$). These results confirm a causal relationship between the enhancement of a firm's digital transformation and its increase in received trade credit and again support Hypothesis H1.

6. Further analyses

6.1. Mechanism tests

Section 5 reveals a causal and positive impact of Chinese firms' digital transformation on their access to trade credit. In this Section, we explore the underlying mechanisms of this impact. As hypothesized in Section 3, we anticipate that digital transformation improves firms' information environment and operational performance, thereby reducing the information asymmetry and default risks for suppliers when providing trade credit. The reduction in risks then strengthens firms' access to trade credit from suppliers.

Table 5
Robustness Tests: Controlling for firm fixed effects.

	(1)	(2)	(3)	(4)
	$TCredit_{i,t+1}$	$TCredit2_{i,t+1}$	$TCredit3_{i,t+1}$	$TCredit4_{i,t+1}$
$DigTran_{i,t}$	0.014***	0.006***	0.012***	0.006***
	(18.64)	(19.17)	(15.34)	(14.98)
ΣControls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
F-statistics	1047.232	781.862	1607.175	1205.097
Adjusted R^2	0.440	0.421	0.475	0.526
# of Obs	25,498	25,498	25,498	25,498

This table reports the estimation results of Eq. (1) in the text while replacing industry fixed effects with firm fixed effects. $DigTran_{i,t}$ is the logarithm of one plus the number of digital transformation-related words mentioned in firm *i*'s annual report of year *t*. $TCredit_{i,t+1}$ is the ratio of firm *i*'s accounts payable to total liabilities at the end of year $t+1$, and $TCredit2_{i,t+1}$ is that of firm *i*'s accounts payable to total assets. $TCredit3_{i,t+1}$ is the ratio of the sum of firm *i*'s accounts payable and notes payable to total liabilities, while $TCredit4_{i,t+1}$ is the ratio of the sum of firm *i*'s accounts payable and notes payable to total assets. ΣControls denotes the set of control variables used in Column (4) of Table 2. We include year fixed effects in all regressions. Parentheses contain t-statistics calculated using robust standard errors. *, **, and *** represent significance at the 10 %, 5 %, and 1 % levels, respectively.

6.1.1. Reducing information asymmetry risk

To empirically verify those mechanisms, we first examine the impact of digital transformation on reducing the firm's information asymmetry risk. Following prior studies, we use analysts' earnings forecast error and dispersion as proxies for a firm's information asymmetry in this study. Numerous studies have pointed out that analysts rely on publicly available information from capital markets to forecast company earnings. The higher the level of information asymmetry, the more difficult it is for analysts to make accurate earnings forecasts or reach a consensus (Armstrong et al., 2014; Boone & White, 2015; Lang et al., 2012). Therefore, analysts' earnings forecast error and dispersion are widely employed as proxies for information asymmetry in prior studies (e.g., de Silva & Thesmar, 2024).

We measure analysts' earnings forecast error ($FError_{i,t}$) as the

Table 6
2SLS regression results based on the instrumental variable approach.

	(1)	(2)
	$DigTran_{i,t}$	$TCredit_{i,t+1}$
$Post_{i,1984} \times Internet_{i,t}$	0.015***	
	(9.86)	
$DigTran_{i,t}$(Instrumented)		0.065***
		(5.66)
ΣControls	Yes	Yes
Industry FE	Yes	Yes
Year FE	Yes	Yes
F-statistic	359.815	282.446
Adjusted R^2/Centered R^2	0.366	0.348
# of Obs	18,801	18,801
Cragg-Donald Wald F statistic	129.562	
Kleibergen-Paap rk Wald F statistic	97.124	

This table reports the two-stage least square regression results based on an instrumental variable. We instrument the firm *i*'s degree of digital transformation ($DigTran_{i,t}$) with the interaction terms of ($Post_{i,1984}$) and ($Internet_{i,t}$). $DigTran_{i,t}$ is the logarithm of one plus the number of digital transformation-related words mentioned in firm *i*'s annual report of year *t*. $Post_{i,1984}$ is the number of post offices per million people in the year 1984 in the city where firm *i*'s headquarter is located. $Internet_{i,t}$ is the number of internet broadband connections in the city where firm *i*'s headquarter is located in year *t*, measured in millions of households. $TCredit_{i,t+1}$ is the ratio of firm *i*'s accounts payable to total liabilities at the end of year $t + 1$. ΣControls denotes the set of control variables used in Column (4) of Table 2. We include industry fixed effects and year fixed effects in all regressions. Parentheses contain t-statistics in Column (1) and z-statistics in Column (2), both of which are calculated with robust standard errors. *, **, and *** represent statistical significance at the 10 %, 5 %, and 1 % levels, respectively.

absolute difference between the mean analyst forecast of firm i's EPS for year t and its actual EPS. Similarly, we calculate $FDisper_{i,t}$ as the standard deviation of analysts' EPS forecasts for firm i in year t. Columns (1) and (2) of Panel A of Table 7 report the results regressing $FError_{i,t}$ and $FDisper_{i,t}$ on $DigTran_{i,t}$, the control variables, and the fixed effects used in Eq. (1), respectively. $DigTran_{i,t}$ is significantly negatively correlated with both $FDisper_{i,t}$ and $FDisper_{i,t}$ at the 5 % level. We then examine the impact of firms' information asymmetry levels on their future access to trade credit ($TCredit_{i,t+1}$) in Columns (1) and (2) of Panel B. Both $FError_{i,t}$ and $FDisper_{i,t}$ are negatively related to $TCredit_{i,t+1}$ with a 10 % or higher significance level. As predicted, these findings support that digital transformation reduces firms' information asymmetry levels, thereby increasing their access to trade credit from suppliers.

6.1.2. Reducing default risk

Next, we verify whether digital transformation enhances firms' ability to obtain trade credit by improving financial solvency and reducing default risk from the perspectives of technical efficiency and financial performance stability. We use firms' total factor productivity ($TFP_{i,t}$) as a proxy for their technical efficiency and the volatility of ROA ($RoaVol_{i,t}$) to measure financial performance stability. Higher $TFP_{i,t}$ indicates that the firm is more efficient in generating output from its inputs, leading to better profitability and stronger cash flow, which in turn supports financial solvency Chen, Harford, and Kamara (2019) and Hasan et al. (2018). Moreover, stable financial performance ensures that a firm has reliable cash inflows, reducing the risk of cash shortfalls that could compromise its ability to meet debt obligations.

We construct $TFP_{i,t}$ using the method of Levinsohn and Petrin (2003), and $RoaVol_{i,t}$ is the standard deviation of the industry-adjusted ROA for firm *i* from year *t*-2 to *t*. Column (3) in Panel A of Table 7 shows that $DigTran_{i,t}$ is significantly positively correlated with $TFP_{i,t}$ at the 1 % level, while Column (4) shows that $DigTran_{i,t}$ is significantly negatively correlated with $RoaVol_{i,t}$ at the 1 % level. Moreover, Columns (3) and (4) in Panel B show that $TFP_{i,t}$ is significantly positively related to $TCredit_{i,t+1}$, while $RoaVol_{i,t}$ is significantly negatively related to $TCredit_{i,t+1}$. These findings support our prediction that digital transformation improves sample firms' technical efficiency and financial performance stability, which in turn enhances the firms' financial solvency and reduces the default risk suppliers face, leading them to provide more trade credit.

Table 7
Mechanism Tests.

Panel A Impact of digital transformation on information asymmetry and financial solvency

	(1)	(2)	(3)	(4)
	$FError_{i,t}$	$FDisper_{i,t}$	$TFP_{i,t}$	$RoaVol_{i,t}$
$DigTran_{i,t}$	−0.016**	−0.008**	0.018***	−0.002***
	(−2.44)	(−2.11)	(3.91)	(−4.64)
ΣControls	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
F-statistics	12.247	85.223	3341.592	36.696
Adjusted R^2	0.101	0.195	0.683	0.045
# of Obs	21,769	19,983	25,240	25,498

Panel B Impact of information asymmetry and financial solvency on trade credit access

	(1)	(2)	(3)	(4)
$DigTran_{i,t}$	$TCredit_{i,t+1}$	$TCredit_{i,t+1}$	$TCredit_{i,t+1}$	$TCredit_{i,t+1}$
$FError_{i,t}$	−0.002*			
	(−2.37)			
$FDisper_{i,t}$		−0.004**		
		(−2.04)		
$TFP_{i,t}$			0.003**	
			(1.97)	
$RoaVol_{i,t}$				−0.019*
				(−1.69)
ΣControls	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
F-statistics	855.857	917.979	986.085	985.942
Adjusted R^2	0.448	0.445	0.432	0.432
# of Obs	21,769	19,983	25,240	25,498

Panel A regresses the firm's analyst forecast error ($FError_{i,t}$), analyst forecast dispersion ($FDisper_{i,t}$), total factor productivity ($TFP_{i,t}$), or ROA volatility ($RoaVol_{i,t}$) on $DigTran_{i,t}$ and control variables. $FError_{i,t}$ is defined as the absolute difference between the mean analyst forecast of firm i's EPS for year t and its actual EPS. $FDisper_{i,t}$ is the standard deviation of analysts' EPS forecasts for firm i in year t. $TFP_{i,t}$ is calculated using the methodology of Levinsohn and Petrin (2003). Finally, $RoaVol_{i,t}$ is the standard deviation of firm *i*'s industry-adjusted ROA from year *t*-2 to year *t*. ΣControls denotes the set of control variables used in Column (4) of Table 2. Panel B examines the impact of $FError_{i,t}$, $FDisper_{i,t}$, $TFP_{i,t}$ and $RoaVol_{i,t}$ on the firm's future access to trade credit ($TCredit_{i,t+1}$). We include industry fixed effects and year fixed effects in all regressions. Parentheses contain t-statistics calculated using robust standard errors. *, **, and *** represent statistical significance at the 10 %, 5 %, and 1 % levels, respectively.

6.2. Heterogeneity analyses

To further verify the mechanisms through which digital transformation impacts firms' access to trade credit, we conduct heterogeneity analyses in this Section to examine whether the impact varies depending on the extent to which firms benefit from digital transformation and the risk of their transformation. Specifically, we use six firm- or region-level cross-sectional characteristics: a firm's size, market share, profitability, and financial risk, as well as the social trust level and law enforcement in the region where the firm is headquartered. Among these, a firm's size and market share determine the benefits of its digital transformation, while profitability and financial risk, along with local social trust level and law enforcement, are related to the risks of digital transformation. Detailed explanations for these predictions can be found in Section 3, where Hypotheses H2 to H4 are developed.

6.2.1. Firm size and market share

As hypothesized in Hypothesis H2, firms with larger sizes and higher market shares possess more economic resources, enabling them to ensure the substantial initial investment required for digital transformation and withstand the risks during the transformation. They are also more likely to benefit from the scale efficiencies arising from the low marginal costs of digital transformation. Therefore, the positive impact of digital transformation on trade credit should be larger for those firms, given that digital transformation is less risky to them and may show greater impacts on improving their operational and financial performance.

To test this hypothesis, we construct dummy variables $DSize_{i,t}$ and $DMShare_{i,t}$. If a sample firm's total assets are above (below) the median of the sample, $DSize_{i,t}$ is assigned a value of 1 (0); similarly, if a firm's market share calculated as its sales revenue as a percentage of total industry revenue is above (below) the sample median, $DMShare_{i,t}$ is assigned a value of 1 (0). We then include the two dummy variables and their interaction terms with $DigTran_{i,t}$ in the baseline model and report the regression results in Columns (1) and (2) of Table 8. In Column (1), the coefficients for $DigTran_{i,t}$ and $DigTran_{i,t} \times DSize_{i,t}$ are 0.012 and 0.004 and are significant at the 1 % level. Those coefficients indicate that the positive impact of digital transformation on trade credit is about (0.004/0.012) = 33.3 % higher in relatively larger firms. Similarly, in Column (2), the coefficients for $DigTran_{i,t}$ and $DigTran_{i,t} \times DSize_{i,t}$ are 0.012 and 0.005, both of which are statistically significant at the 1 % level. Those results confirm that the effect of digital transformation on enhancing firms' ability to acquire trade credit is nearly 40 % more in firms with higher market shares. In summary, the results in Columns (1) and (2) of Table 8 validate our Hypothesis H2.

6.2.2. Profitability and financial risk

Like any other strategic transformation, digital transformation is risky since it is costly and has highly uncertain outcomes (Chen & Srinivasan, 2024). This transformational risk may compound with the firm's operational or financial risks, weakening its ability and willingness to fulfill debt obligations. Consequently, the higher a firm's existing operational and financial risks, the more cautious its suppliers might be regarding its digital transformation. Based on this assertion, Hypothesis H3 anticipates that the positive impact of digital transformation on trade credit is stronger in firms with higher profitability and lower bankruptcy risk. The reason is that firms' profitability negatively correlates with their operational and financial risk, while firms' bankruptcy risk directly indicates their financial distress and default risk.

To test Hypothesis H3, we use ROE to measure a firm's profitability and the *Z*-Score of Altman (1968) as an indicator of bankruptcy risk, where a higher Z-Score indicates lower bankruptcy likelihood and financial risk. We then construct two dummy variables. The dummy $DProfit_{i,t}$ is assigned a value of 1 if a sample firm's ROE exceeds the median ROE of its industry in the same year and 0 otherwise; similarly, $DZscore_{i,t}$ equals 1 if a firm's Z-Score is above the sample median and 0 otherwise. We then include the two dummies, along with their interaction terms with $DigTran_{i,t}$, in the baseline regression model. Columns (3) and (4) of Table 8 show that $DigTran_{i,t}$ and both interaction terms are positively correlated with $TCredit_{i,t+1}$ with a 10 % or higher significance level. These results support Hypothesis H3 and suggest that the enhancement effect of digital transformation on a firm's ability to obtain

Table 8
Heterogeneity Analyses.

Dep Var = $TCredit_{i,t+1}$	(1)	(2)	(3)	(4)	(5)	(6)
$DigTran_{i,t}$	0.012***	0.012***	0.016***	0.015***	0.015***	0.015***
	(11.29)	(12.14)	(16.09)	(16.35)	(16.46)	(15.85)
$DigTran_{i,t} \times DSize_{i,t}$	0.004***					
	(3.64)					
$DSize_{i,t}$	−0.011***					
	(−4.32)					
$DigTran_{i,t} \times DMShare_{i,t}$		0.005***				
		(3.82)				
$DMShare_{i,t}$		0.007***				
		(2.89)				
$DigTran_{i,t} \times DProfit_{i,t}$			0.003**			
			(2.24)			
$DProfit_{i,t}$			0.010***			
			(4.45)			
$DigTran_{i,t} \times DZscore_{i,t}$				0.002*		
				(1.85)		
$DZscore_{i,t}$				0.029***		
				(11.33)		
$DigTran_{i,t} \times DTrust_{i,t}$					−0.002**	
					(−2.01)	
$DTrust_{i,t}$					0.011***	
					(5.93)	
$DigTran_{i,t} \times DLaw_{i,t}$						−0.003***
						(−2.68)
$DLaw_{i,t}$						0.013***
						(6.62)
ΣControls	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
F-statistics	889.167	889.046	889.369	903.275	880.691	892.828
Adjusted R^2	0.440	0.441	0.440	0.443	0.445	0.442
# of Obs	25,498	25,498	25,498	25,498	24,824	25,070

This table reports the results examining how the impact of digital transformation on trade credit varies depending on firms' size, market share, profitability, financial distress, and local social trust levels or law enforcement intensity. $DSize_{i,t}$ equal 1 if firm *i*'s size measured by its total assets is above the sample median and 0 otherwise. $DMShare_{i,t}$ equal 1 if firm *i*'s sales as a proportion of total industry sales are higher than the sample median and 0 otherwise. $DProfit_{i,t}$ ($DZscore_{i,t}$) equal 1 if firm *i*'s ROE (financial distress risk proxied by Altman's *Z*-score) is above the sample median and 0 otherwise. Finally, $DTrust_{i,t}$ ($DLaw_{i,t}$) equals 1 for firms headquartered in provinces with above-median social trust levels (law enforcement intensity). ΣControls denotes the set of control variables used in Column (4) of Table 2. We include year fixed effects and industry fixed effects in all regressions. Parentheses contain t-statistics calculated using robust standard errors. *, **, and *** represent significance at the 10 %, 5 %, and 1 % levels, respectively.

trade credit is significantly stronger in firms with higher profitability and lower bankruptcy risk.

6.2.3. Local social trust level and law enforcement

Finally, we predict that the impact of digital transformation on trade credit varies across different regions. First, in regions with higher social trust levels, the trust between suppliers and customer firms is greater (Wu et al., 2014), thereby diminishing the marginal effect of digital transformation in enhancing the firm's trustworthiness. Second, the interests of suppliers as creditors are better legally protected in regions with stronger law enforcement. Formal legal protection may substitute for the enhancement of a firm's credibility through digital transformation. Based on these assertions, Hypothesis H4 posits that the positive impact of digital transformation on trade credit is weaker in regions with higher social trust levels and stronger law enforcement.

To test Hypothesis H4, we follow Wu et al. (2014) and construct a Chinese provincial-level social trust index using data from the 2015 Chinese General Social Survey (CGSS). CGSS asks local respondents how much they trust strangers, with answers ranging from "very distrustful" to "very trustworthy" on a five-point scale (1 to 5). The average score of respondents in each province represents the overall social trust level in that province. Based on this assertion, we use data from the 2015 CGSS survey to construct social trust proxies for the entire sample period. This approach is appropriate because regional social trust levels are relatively stable over time and exhibit no significant variation across different periods (Wu et al., 2014). Using the CGSS trust scores, we construct a dummy variable $DTrust_{i,t}$, which is set to 1 if the province where firm i's headquarter is located has a social trust level above the sample median and 0 otherwise.

Additionally, we measure the law enforcement intensity in each province of China using the "Market Intermediaries Development and Legal Environment" sub-index from the "China Provincial Marketization Index Reports."[11] The dummy variable $DLaw_{i,t}$ is set to 1 if the above-mentioned sub-index of the province where firm i's headquarter is located is above the sample median and 0 otherwise.

Columns (5) and (6) of Table 8 include $DTrust_{i,t}$, $DLaw_{i,t}$, and their interaction terms with $DigTran_{i,t}$ in the regressions. The coefficient for $DigTran_{i,t} \times DTrust_{i,t}$ is −0.002 with a 5 % significance level, indicating that the positive impact of digital transformation on trade credit is significantly weaker in provinces with higher levels of social trust. Similarly, the coefficient for $DigTran_{i,t} \times DLaw_{i,t}$ is negative with a 1 % significance level, confirming that suppliers in regions with stronger law enforcement care less about firms' digital transformation when providing trade credit. In summary, Columns (5) and (6) of Table 8 support Hypothesis H4.

6.3. Economic consequences

The previous sections confirm that a firm's digital transformation can aid in securing more trade credit from suppliers, and this effect exhibits considerable heterogeneity. In this Section, we investigate the impact of trade credit a firm receives on its financing constraints, debt costs, and investment efficiency, thereby demonstrating the economic consequences of the impact of digital transformation on the firm's ability to acquire trade credit.

Firstly, trade credit is an informal form of short-term financing suppliers provide. Therefore, more available trade credit for a firm may help alleviate its financing constraints.[12] To test this prediction, we employ the SA index proposed by Hadlock and Pierce (2010) to measure the extent of a firm's financing constraints ($SA_{i,t}$), with a larger SA index indicating lighter financing constraints. As shown in Column (1) of Table 9, where we regress $SA_{i,t}$ on $TCredit_{i,t}$ and control variables, the coefficient of $TCredit_{i,t}$ is positive (0.084) with a 1 % significance level. As predicted, more available trade credit significantly eases the firm's financing constraints.

Secondly, we predict that increased availability of trade credit reduces a firm's reliance on other forms of debt capital (such as bank loans), enhancing its bargaining power with potential creditors and reducing its debt costs. To test this prediction, we use the ratio of the firm's financial expenses to total liabilities to measure its debt financing costs and denoted it as $DCost_{i,t}$. Column (2) of Table 9 shows that the trade credit proxy, $TCredit_{i,t}$, is significantly negatively correlated with $DCost_{i,t}$, supporting our prediction that firms with more access to trade credit incur lower debt costs.

Finally, we expect that more available trade credit will decrease the firm's investment inefficiency. On the one hand, as previously discussed and empirically verified, trade credit access can alleviate firms' financing constraints. Consequently, more available trade credit eases firms' potential underinvestment problems. On the other hand, like other creditors, suppliers have motives and channels to proactively govern their customer firms to protect their interests (Hui et al., 2012). Such governance efforts may restrain managers' opportunistic tendencies towards empire-building, thereby curbing their overinvestment. In summary, trade credit may mitigate both firms' underinvestment and overinvestment, thereby enhancing their investment efficiency.

Table 9
Economic Consequences.

	(1)	(2)	(3)
	$SA_{i,t}$	$DCost_{i,t}$	$IneffIn_{i,t}$
$TCredit_{i,t}$	0.084***	−0.873***	−0.043***
	(6.57)	(−4.93)	(−5.10)
ΣControls	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
F-statistics	125.855	1027.669	35.554
Adjusted R^2	0.236	0.533	0.066
# of Obs	24,801	25,481	21,830

This table demonstrates the economic consequences of the impact of digital transformation on trade credit by examining how firms' access to trade credit affects their financing constraints, debt costs, and investment inefficiency. $TCredit_{i,t}$ is the ratio of firm i's accounts payable to total liabilities at the end of year t. $SA_{i,t}$ is the SA index that proxies for firm i's financing constraints and is calculated in the way of Hadlock and Pierce (2010). $DCost_{i,t}$ is the ratio of firm i's financial expenses to total liabilities in year t, and $IneffIn_{i,t}$ is firm i's investment inefficiency calculated in the way of Richardson (2006). ΣControls denotes the set of control variables used in Column (4) of Table 2. We include year fixed effects and industry fixed effects in all regressions. Parentheses contain t-statistics calculated using standard errors clustered at the firm level. *, **, and *** represent significance at the 10 %, 5 %, and 1 % levels, respectively.

[11] These reports are compiled by Fan Gang from the China Economic Reform Institute, Wang Xiaolu from the National Development Research Institute of Peking University, along with other scholars in collaboration. The index includes sub-indices such as Government and Market Relationship, Non-State Sector Development, Product Market Development, Factor Market Development, Legal and Institutional Environment, and Market Intermediaries and Social Organizations. Both the index and its sub-indices are widely employed in studies on economic and financial topics in China (e.g., Ke and Zhang (2021) and Gao et al. (2023)).

[12] The relationship between firms' access to trade credit and their financial constraints or financing costs is not straightforward. The substitutionary financing theory, argue that financially constrained firms rely more on trade credit as an alternative to limited formal financing (Levine et al., 2018; Chen et al. 2019b). Conversely, the transaction motivation theory posits that trade credit is provided by suppliers to enhance competitiveness and build customer relationships, making its use unrelated to a firm's financial constraints (Fabbri & Menichini, 2010).

To test this prediction, we employ the method of Richardson (2006) to construct the proxy for firms' investment inefficiency and denote it as $IneffIn_{i,t}$. By definition, a larger $IneffIn_{i,t}$ indicates lower investment efficiency, i.e., greater underinvestment or overinvestment. As shown in Column (3) of Table 9, $TCredit_{i,t}$ is significantly negatively correlated with $IneffIn_{i,t}$ at the 1 % significance level, indicating that firms receiving more trade credit show significantly higher investment efficiency.

In summary, while previous sections prove that Chinese firms with higher degrees of digital transformation receive more trade credit from suppliers, the results in Table 9 suggest that increased trade credit helps firms ease financing constraints, reduce debt financing costs, and enhance investment efficiency. These findings not only reveal the economic consequences of the impact of digital transformation on firms' access to trade credit but also contribute to understanding the mechanisms through which firms' digital transformation improves their operational and financial performance.

7. Conclusions

Using a sample of Chinese public firms, we explore how suppliers respond to customer firms' digital transformation by examining the impact of such transformation on firms' access to trade credit granted by suppliers. Our empirical results demonstrate a significant and robust positive correlation between the degree of a firm's digital transformation and the quantity of trade credit it receives. In terms of the mechanisms, we find that digital transformation improves firms' information environment, reducing the information asymmetry risk faced by suppliers. It also helps improve firms' financial solvency by increasing their operational efficiency and ROA stability. Both the decrease in information asymmetry risk and the improvement in financial solvency enhance firms' access to trade credit from suppliers. We also demonstrate the economic importance of the positive impact by showing that receiving more trade credit from suppliers alleviates firms' financing constraints, reduces their debt costs, and improves their investment efficiency.

In summary, we establish the link between firms' digital transformation and their access to trade credit. By doing so, we add to the studies on the outcomes of corporate digital transformation and those on the determinants of suppliers' trade credit provision. Additionally, our findings regarding the economic consequences of increased trade credit offer valuable insights into understanding the mechanisms through which digital transformation improves firms' operational or financial performance.

Appendix A. Variable definitions

Variable	Definitions
$TCredit_{i,t+1}$	The ratio of firm i's accounts payable to total liabilities at the end of year $t+1$.
$TCredit2_{i,t+1}$	The ratio of firm i's accounts payable to total assets at the end of year $t+1$.
$TCredit3_{i,t+1}$	The ratio of the sum of firm i's accounts payable and notes payable to total liabilities at the end of year $t+1$.
$TCredit4_{i,t+1}$	The ratio of the sum of firm i's accounts payable and notes payable to total assets at the end of year $t+1$.
$DigTran_{i,t}$	The logarithm of one plus the number of digital transformation-related feature words mentioned in firm i's annual report of year t.
$Size_{i,t}$	The logarithm of firm i's total assets at the end of year t, measured in billion yuan.
$BM_{i,t}$	The book-to-market ratio of firm i at the end of year t.
$SOE_{i,t}$	A dummy variable where state-owned enterprises are assigned a value of 1, and non-state-owned enterprises are assigned a value of 0.
$BCredit_{i,t}$	The ratio of the sum of firm i's short-term and long-term borrowings to total liabilities at the end of year t.
$SDebt_{i,t}$	The ratio of firm i's current liabilities, excluding accounts payable, to total liabilities at the end of year t.
$Lev_{i,t}$	The debt-to-asset ratio of firm i at the end of year t.
$Cash_{i,t}$	The ratio of the sum of firm i's cash and cash equivalents and net short-term investments to total assets at the end of year t.
$Tan_{i,t}$	The ratio of firm i's net tangible assets to total assets at the end of year t.
$ROA_{i,t}$	The return on assets (ROA) of firm i at the end of year t.
$SaleGr_{i,t}$	The growth of firm i's sales revenue in year t relative to the previous year.
$Internet_{i,t}$	The number of internet broadband connections in the city where firm i's headquarter is located in year t, measured in millions of households.
$Post_{i,1984}$	The number of post offices per million people in the year 1984 in the city where firm i's headquarter is located.
$FError_{i,t}$	The absolute difference between the mean analyst forecast of firm i's EPS for year t and its actual EPS.
$FDisper_{i,t}$	The standard deviation of analysts' EPS forecasts for firm i in year t.
$TFP_{i,t}$	The total factor productivity of firm i in year t, measured in the methodology of Levinsohn and Petrin (2003).
$RoaVol_{i,t}$	The standard deviation of firm i's industry-adjusted ROA from year t-2 to year t.
$SA_{i,t}$	The financing constraints for firm i in year t, measured by the SA index using the methodology of Hadlock and Pierce (2010).
$DCost_{i,t}$	The ratio of firm i's financial expenses to total liabilities in year t.
$IneffIn_{,t}$	The extent of inefficient investment for firm i in year t, calculated using the Richardson (2006) model.

Appendix B. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.irfa.2025.103976.

Data availability

Data will be made available on request.

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